

Question 12

- Machine Learning (ML) is a part of artificial intelligence that allows computers to learn from data and improve its performance over time without being directly programmed.
- It differs from traditional programming by that in traditional programming a human developer writes the rules whereas in ML the computer discovers them from data.

• Types of ML

① Supervised ML

This is a type of ML where a computer is trained using labeled data. For every input, the output is already known. For example a machine is given features of a bird and asked to predict after learning the features.

② Unsupervised ML

This is a type of ML where a machine is trained without labeled data. The machine does not know the answer beforehand. It learns by finding patterns, similarities and differences in the given data. For example asking a machine to categorize a basket full of different fruits.

③ Reinforcement ML

This is whereby a machine learns from experience through the feedback process. It is rewarded for each good action and penalized for every wrong move. It is applied for example in gaming and multi-agent systems.